

Software Implementation Manual

Feature Reference Guide for Stress Level Assessment System

This document provides technical specifications for implementing the **Stress Level Dataset** into a software application. It outlines how each feature should be presented in the User Interface (UI), the allowed value ranges, and a brief description for tooltips or labels.

1. Psychological Factors

Feature Name	Recommended UI	Value Range	Description & Logic
Anxiety Level anxiety_level	Number Input / Slider	0 to 21	Standard anxiety score. Higher score means higher anxiety.
Depression depression	Number Input / Slider	0 to 27	Standard depression score. Higher score indicates severe depression.
Self Esteem self_esteem	Number Input / Slider	0 to 30	Measures personal worth. Lower score often correlates with higher stress.
Mental Health History mental_health_history	Radio / Toggle	0 or 1	0 = No previous history 1 = Has previous history

2. Physical & Health Factors

Feature Name	Recommended UI	Value Range	Description & Logic
Blood Pressure blood_pressure	Dropdown	1, 2, 3	1 = Low BP 2 = Normal BP 3 = High BP
Sleep Quality sleep_quality	Star Rating / Slider	0 to 5	0 = Very Poor Sleep 5 = Excellent Sleep
Headache headache	Rating Scale	0 to 5	0 = No Headaches 5 = Severe/Frequent Headaches
	Rating Scale	0 to 5	

Breathing Problem breathing_problem			0 = No Issues 5 = Severe Breathing Issues
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3. Environmental Factors

Feature Name	Recommended UI	Value Range	Description & Logic
Noise Level noise_level	Rating Scale	0 to 5	0 = Quiet Environment 5 = Highly Noisy Environment
Living Conditions living_conditions	Rating Scale	0 to 5	0 = Very Poor 5 = Excellent/Comfortable
Safety safety	Rating Scale	0 to 5	0 = Feels Unsafe 5 = Feels Completely Safe
Basic Needs basic_needs	Rating Scale	0 to 5	0 = Needs Unmet 5 = Needs Fully Met

4. Academic & Social Factors

Feature Name	Recommended UI	Value Range	Description & Logic
Academic Performance academic_performance	Rating Scale	0 to 5	0 = Failing/Poor 5 = Excellent Performance
Study Load study_load	Rating Scale	0 to 5	0 = No Load 5 = Extremely High Load
Future Career Concerns future_career_concerns	Rating Scale	0 to 5	0 = No Concern 5 = Highly Concerned
Peer Pressure & Bullying peer_pressure / bullying	Rating Scale	0 to 5	0 = None 5 = Severe Pressure/ Bullying
Social Support social_support	Rating Scale	0 to 3	0 = No Support 3 = Strong Support System

5. Target Variable (Model Output)

Feature Name	Recommended UI	Value Range	Description & Logic
Stress Level stress_level	Read-Only / Dashboard Widget	0, 1, 2	Output generated by the Machine Learning Model. 0 = Low Stress 1 = Medium Stress 2 = High Stress